Lecture - 9

We claimed that the Converse of Lagrange's theorem is not true in general. But we will prove that Converse of Lagrange's theorem is true for finite cyclic groups.

Theorem: Let G be a finite cyclic group of order n. Let d be a positive integer such that $d \mid n$. Then G has a subgroup of order d.

Proof: Since G is a cyclic group, $\exists\, g \in G$ such that

$$G = \langle g \rangle = \{ g^k \mid k \in \mathbb{Z} \}.$$

as $|G| = n$, we have $|g| = n$.

So $$|g^{\frac{n}{d}}| = \frac{n}{\gcd(n, \frac{n}{d})} = \frac{n}{n/d} = d.$$

So the order of the element $g^{\frac{n}{d}}$ is d.

Define $H = \langle g^{n/d} \rangle$ i.e. the subgroup generated by $g^{\frac{n}{d}}$

So $$H = \{ (g^{n/d})^k \mid k \in \mathbb{Z} \}.$$

Since $|g^{n/d}| = d$.

we have $$H = \{ 1, g^{n/d}, (g^{n/d})^2, (g^{n/d})^3, \ldots, (g^{n/d})^{d-1} \}.$$

So $|H| = d$. Hence the claim.

<u>Example:</u> Consider $(\mathbb{Z}_8, +)$.

$|\mathbb{Z}_8| = 8$. & $\mathbb{Z}_8$ is generated by $[1]$.

$2 \mid 8$ so $8/2\,[1] = [4]$ generates a subgroup of order 2.

$$H = \langle [4] \rangle = \{[4], [0]\}$$

Similarly $4 \mid 8$. so $8/4\,[1] = [2]$ generates a subgroup of order 4.

$$H' = \langle [2] \rangle = \{[0], [2], [4], [6]\}.$$

<u>Note:</u> Every cyclic group is abelian.

Let G be cyclic. Then $\exists\, g \in G$ s.t. $G = \langle g \rangle = \{g^k \mid k \in \mathbb{Z}\}$.

Let $a, b \in G$.

Then $a = g^{k_1}$ for some $k_1 \in \mathbb{Z}$

$b = g^{k_2}$ for some $k_2 \in \mathbb{Z}$.

So $a \cdot b = g^{k_1} \cdot g^{k_2} = g^{k_1 + k_2} = b \cdot a$.

Hence G is abelian

Classification of Subgroups of cyclic groups

① Every subgroup of a cyclic group is cyclic.

<u>Proof:</u> Let G be a cyclic group. i.e. $\exists\, g \in G$ such that

$$G = \langle g \rangle = \{ g^n \mid n \in \mathbb{Z} \}.$$

Let $H \leq G$. Need to prove H is cyclic.

If $H = \{e\}$ then $H = \langle e \rangle$.

Suppose $H \neq \{e\}$. So $\exists\, k$ non-zero integer s.t.

$g^k \in H$.

Since H is a subgroup, $(g^k)^{-1} \in H \Rightarrow g^{-k} \in H$

Either k or $-k$ is a positive integer.

$\Rightarrow \exists$ a positive integer l s.t. $g^l \in H$.

Consider $S = \{ m \in \mathbb{N} \mid g^m \in H \}$.

By previous argument $S \neq \phi$

By well-ordering principle, the set S has a smallest element, say α.

i.e. $g^\alpha \in H \Rightarrow \langle g^\alpha \rangle = \{ e, g^\alpha, g^{2\alpha}, \dots \} \subseteq H$.

<u>Claim:</u> $\langle g^\alpha \rangle = H$.

Let $x \in H$ so $x = g^\beta$ for some $\beta \in \mathbb{Z}$.

By division algorithm $\exists !\ q$ & r such that

$\beta = q\alpha + r \qquad 0 \leq r < \alpha$

$\Rightarrow x = g^\beta = g^{q\alpha} \cdot g^r \quad \therefore g^r = (g^{q\alpha})^{-1} x \in H$

as $g^\alpha, x \in H$.

As $0 \le r < \alpha$, & ~~g^α is~~ α is smallest positive integer such that $g^r \in H$, we must have

$$r = 0.$$

$$\Rightarrow \quad \beta = g^{q\alpha}$$

$$\Rightarrow \quad x = g^{\beta} = g^{q\alpha} \in \langle g^{\alpha} \rangle$$

So $H = \langle g^{\alpha} \rangle$. Hence H is cyclic.

(2) If $G = \langle g \rangle$ is a cyclic group & $H \le G$ then $|H| \mid |G|$. Conversely if d is a divisor of $|G|$, then there is a subgroup $\langle g^{\frac{|G|}{d}} \rangle$, whose order is d.

(3) If $G = \langle g \rangle$ cyclic & $g \in G$, then $|g| \mid |G|$.

Since $g \in G$, Consider the subgroup generated by g,

$$\langle g \rangle = \{ g^n \mid n \in \mathbb{Z} \}.$$

Check (1) $\langle g \rangle \le G$.

(2) $|\langle g \rangle| = |g|$.

By Lagrange's theorem $|\langle g \rangle| = |g| \mid |G|$.

Let G be a group & $H, K \leq G$. Define

$$HK = \{hk \mid h \in H, k \in K\}.$$

Q. When is HK a subgroup of G?

<u>Theorem:</u> $HK \leq G$ if and only if $HK = KH$

<u>Proof:</u> Let $HK \leq G$.

Claim $HK = KH$.

Let $x \in HK$ then $x = hk$ for some $h \in H$ & $k \in K$.

Since $HK \leq G$, $x^{-1} = (hk)^{-1} = k^{-1}h^{-1} \in HK$.

$\Rightarrow$ $x^{-1} = h_1 k_1$ for some $h_1 \in H$ & $k_1 \in K$.

$x = (x^{-1})^{-1} = (h_1 k_1)^{-1} = k_1^{-1} h_1^{-1} \in KH$.

$\Rightarrow$ $HK \subseteq KH$.

Similarly $KH \subseteq HK$. $\Rightarrow$ $HK = KH$.

Conversely, Suppose $HK = KH$.

Claim $HK \leq G$.

Let 1 be the identity in G. $1 \in H$ & $1 \in K$.

$\Rightarrow$ $1 = 1 \cdot 1 \in HK$.

Let $a, b \in HK$.

So $a = h_1 k_1$ for some $h_1 \in H$, $k_1 \in K$

$b = h_2 k_2$ for some $h_2 \in H$ $k_2 \in K$.

Need to show

$$a^{-1}b = (h_1 k_1)^{-1} h_2 k_2 \in HK.$$

$a^{-1}b = (h_1k_1)^{-1} h_2k_2$

$= k_1^{-1}h_1^{-1}h_2k_2$

$k_1^{-1} \in K \quad h_1^{-1}h_2 \in H. \quad [\because H \leq G]$

$\Rightarrow k_1^{-1}h_1^{-1}h_2 \in KH = HK.$

$\Rightarrow k_1^{-1}h_1^{-1}h_2 = h_3k_3$ for some $h_3 \in H$, $k_3 \in K$.

$\Rightarrow a^{-1}b = h_3k_3k_2$

$k_3k_2 \in K \quad [\because K \leq G]$.

$\Rightarrow a^{-1}b \in HK.$ Hence $HK \leq G$.

Corollary: If H & K are subgroups of a group G such that either $H \trianglelefteq G$ or $K \trianglelefteq G$, then $HK \leq G$.

Proof: To show $HK \leq G$, it is enough to prove

$HK = KH.$

Let $H \trianglelefteq G$. then $gH = Hg \quad \forall g \in G$.

in particular $kH = Hk \quad \forall k \in K$.

$$HK = \bigcup_{k \in K} Hk = \bigcup_{k \in K} kH = KH.$$

Hence $HK \leq G$.

Similarly for $K \trianglelefteq G$, we get $HK \leq G$.

Theorem: If H, K are finite subgroups of a group G, then

$$|KH| = |HK| = \frac{|H| \cdot |K|}{|H \cap K|}$$

Proof: $K \leq G$, $H \leq G$. So $H \cap K \leq G$.

Since $H \cap K \subseteq K$, $H \cap K \leq K$.

Since K is finite $[K : H \cap K] = \dfrac{|K|}{|H \cap K|}$ is also finite.

$t = [K : H \cap K]$ is the number of distinct left cosets of $H \cap K$ in K.

Let $a_1(H \cap K), a_2(H \cap K), \ldots, a_t(H \cap K)$ be all the distinct left cosets of $H \cap K$ in K.

So $K = a_1(H \cap K) \cup a_2(H \cap K) \cup \cdots \cup a_t(H \cap K)$

So $KH = \bigcup\limits_{k \in K} kH = a_1(H \cap K)H \cup a_2(H \cap K)H \cup \cdots \cup a_t(H \cap K)H$

$= a_1H \cup a_2H \cup \cdots \cup a_tH.$

$(H \cap K)H = H$ as $H \cap K \subseteq H$. & $H \leq G$.

Claim $a_iH = a_jH$ if and only if $i = j$.

where $1 \leq i, j \leq t$.

$a_iH = a_jH \Leftrightarrow a_i^{-1}a_j \in H$.

By definition $a_i, a_j \in K \Rightarrow a_i^{-1}a_j \in K$ $[\because K \leq G]$

$\Leftrightarrow a_i^{-1}a_j \in H \cap K$.

$\Leftrightarrow a_i(H \cap K) = a_j(H \cap K) \Leftrightarrow i = j$.

So $KH = a_1 H \cup a_2 H \cup \cdots \cup a_t H$ 'disjoint union of left cosets of H in G.

$$\therefore |KH| = \sum_{i=1}^{t} |a_i H|$$

$$= \sum_{i=1}^{t} |H| = t \cdot |H| = \frac{|K|}{|H \cap K|} \cdot |H|.$$

Similarly by considering the right cosets of $H \cap K$ in K, we can prove $|HK| = \dfrac{|H| \cdot |K|}{|H \cap K|}$.

Hence the result.

<u>Quotient group:</u> Let G be a group & $H \leq G$.

Consider G/H = set of all left cosets of H in G.

<u>Goal:</u> To define a binary operation on G/H under which G/H becomes a group.

Given two cosets $g_1 H$ & $g_2 H$, a natural way to define an operation on them is to define

$$(g_1 H)(g_2 H) = (g_1 g_2) H.$$

But we need to check that this operation is well defined.

i.e. if $g_1 H = g_1' H$ & $g_2 H = g_2' H$ then

$g_1 g_2 H$ should be same as $g_1' g_2' H$.